ODE HW7

漏电导. 沿轴突传播的扩散项

$$C\frac{\partial V}{\partial t}+gV=\underline{\frac{r}{2\rho}\frac{\partial^2 V}{\partial x^2}}+I_{Na}m(1-n)-I_K n+I_0$$

膜电容 (电压变化的惯性) 钠电流

$$\tau\frac{\partial n}{\partial t}=m-n$$

$$m=\begin{cases}0 & V\le V^* \Rightarrow \text{钠通道关}\\ 1 & V>V^* \ \text{钠通道开}\end{cases}$$

n变化的时间尺度. τ越大, n反应越慢.
n会慢慢追着m跑. τ是时间常数.
m=1, 右边 $1-n>0$: n慢慢往1靠
m=0, 右边 $-n<0$: n慢慢往0靠

这是一根神经纤维的电缆方程，它告诉你沿着轴突的位置x，在时间t上，电压 $V(x,t)$ 怎么随时间变化；同时 $n(x,t)$ 是一个恢复/通道开关的变量，决定离子电流大小，外面有恒定外加电流 I_0 在刺激它。

$V(x,t)$ 膜电位(电压) mV

$n(x,t)$: 介于0和1之间的门控变量，也就是钾离子通道打开的程度. (0全关, 1全开)

1. Constant solution，既不随时间变，也不随空间变的解.

$\frac{\partial V}{\partial t}=0 \quad \frac{\partial^2 V}{\partial x^2}=0 \quad \frac{\partial n}{\partial t}=0$. 这时, V, n 都是 constant.

代入PDE: $gV=I_{Na}m(1-n)-I_K n+I_0$

$0=m-n \Rightarrow m=n.$

① If $m=0$. $\tau n_t=m-n$ $m=n=0$.
$gV=I_0 \Rightarrow V=\frac{I_0}{g}$
$\Downarrow$
$V\le V^* \Rightarrow \frac{I_0}{g}\le V^*$

② If $m=1$ $m=n=1$
$gV=I_{Na}(1-1)-I_K\cdot 1+I_0$
$V=\frac{I_0-I_K}{g}$
$V>V^*$ $\frac{I_0-I_K}{g}>V^*$

Thus, when $I_0\le gV^*$, there is one sol $V=\frac{I_0}{g}$ $m=0$ $n=0$

$I_0>gV^*+I_K$ there is one sol $V=\frac{I_0-I_K}{g}$ $n=1$ $m=1$.

$gV^*\le I_0\le gV^*+I_K$, there is no sol.

2. Periodic sols independent of x.

找个周期性放电-恢复过程.
像心跳/神经冲动.

$\frac{\partial^2 V}{\partial x^2}=0$. 先不管空间，只看时间. PDE ⇒ ODE

$CV'(t)+gV(t)=I_{Na}m(1-n)-I_K n+I_0$

$\tau n'(t)=m-n.$

$[-T_0,0]$: $V(t)\le V^*$, $m=0 \Rightarrow \tau n'=-n$ $n(0)=n_0\Rightarrow n(t)=n_0e^{-t/\tau}$

$(0,T_1)$: $V(t)>V^*$, $m=1 \Rightarrow \tau n'=1-n$ $n(0)=n_0\Rightarrow n(t)=1-(1-n_0)e^{-t/\tau}$

$n(-T_0)=n(T_1)=n_1$

$T_0=\tau\log\frac{n_1}{n_0}$ $T_1=\tau\log\frac{1-n_0}{1-n_1}$

$CV'(t)+gV(t)=-I_K n(t)+I_0=-I_K n_0e^{-t/\tau}+I_0$ $-T_0\le t\le 0$

$V'(t)+\frac{g}{C}V(t)=\frac{I_0}{C}-\frac{I_K n_0}{C}e^{-t/\tau}$

我们要把V(t)写成特解+齐次解 $CV_h'+gV_h=0 \Rightarrow V_h(t)=Ae^{-(g/c)t}$

特解

$V_p(t) = A_0 + B_0 e^{-t/\tau}$

plug V_p back. solve A_0, B_0.

$V(t) = V_p(t) + V_h(t) = A_0 + B_0 e^{-t/\tau} + A e^{-(g/c)t}$

$V(-T_0) = V^*$ we solved $A = \frac{I_0}{g}$ $B_0 = -\frac{I_k n_1}{g - \frac{c}{\tau}}$

$V(t) = A_0 + B_0 e^{-(t+T_0)/\tau} + (V^* - A_0 - B_0) e^{-(g/c)(t+T_0)}$ $\quad -T_0 \le t \le 0$

$(0, T_1]$ $m = 1$.

$n(t) = 1-(1-n_0)e^{-t/\tau}$

$CV'(t) + gV(t) = I_{Na}(1-n) - I_K n + I_0$

$= (I_0 - I_K) + (I_{Na}+I_K)(1-n_0)e^{-t/\tau}$

$V_p^+(t) = A_1 + B_1 e^{-t/\tau}$

$V_+(t) = V_p^+(t) + k_+ e^{-(g/c)t}$ $\quad A_1 = \frac{I_0 - I_K}{g}$ $\quad B_1 = \frac{(I_{Na}+I_K)(1-n_0)}{g-\frac{c}{\tau}}$

$V^* = V_+(0) = A_1 + B_1 + k_+ \Rightarrow k_+ = V^* - A_1 - B_1$

$V_+(t) = A_1 + B_1 e^{-t/\tau} + (V^* - A_1 - B_1)e^{-(g/c)t}$ $\quad 0 \le t \le T_1$

endpoint

$[-T_0, 0]$ end $\quad V_-(0) = V^*$

$t=0$ $\quad V_-(0; n_0, n_1, T_0) = V^*$ $\quad T_0 = \tau \log \frac{n_1}{n_0}$

$(0, T_1]$ end. $\quad V_+(T_1) = V^*$

$t = T_1$ $\quad V_+(T_1; n_0; n_1, T_1) = V^*$ $\quad T_1 = \tau \log \frac{1-n_0}{1-n_1}$

Period I_0 Image

Fixing the HH parameters at the values given in the assignment, I solved the exact equation for (n_0, n_1) numerically for each $I_0 \in (20, 50)$, and computed the period $T = T_0 + T_1$.
I also repeated the computation with a very small capacitance C to approximate the limit $C \to 0$.
The resulting curves $T(I_0)$ are shown in Figure I. In both the exact and approximate cases the period is a decreasing function of I_0: stronger input current leads to more frequent firing.
The $C \to 0$ approximation slightly underestimates the period but captures the overall shape very well.

$V(t)$ images

Figure 2-4 show the corresponding space-independent periodic solution $V(t)$ for three representative values of I_0 in the allowed interval. In each case $V(t)$ is continuous and periodic, and crosses the threshold V^* twice every period, as required by the construction. For I_0 just above 20 the upstroke is a very sharp and the downstroke is long; as I_0 increases, the spike becomes shorter in time and the period decreases. When I_0 is closed to 50, the oscillation has small amplitude around V^*, consistent with the appearance of $m=1$ steady state at $I_0 = 50$.

3. $V(x,t) = V(T)$ $n(x,t) = N(T)$ $m(x,t) = m(T)$, $T = t + \frac{x}{w}$

By chain rule, we obtained
$$\begin{cases} \frac{-r}{2\rho w^2} V'' + cV' + gV = I_{Na} M(1-N) - I_k N + I_0 \\ rN' = M - N \\ M = \begin{cases} 0 & V \le V* \\ 1 & V > V* \end{cases} \end{cases}$$

It's the same as we did in (2).

$$N(T) = N_1 e^{\left(-\frac{T+T_0}{r}\right)} \quad -T_0 \le T \le 0$$

$$1 - N(T) = (1-N_0) e^{\left(-\frac{T}{r}\right)} \quad 0 < T < T_1 \quad , N_0 = N(0) \; N_1 = N(T_1) = N(-T_0)$$

$$T_0 = r \log \frac{N_1}{N_0} \quad T_1 = r \log \frac{1-N_0}{1-N_1}$$

Now we consider $-\frac{r}{2\rho w^2} V'' + cV' + gV = 0$

we try $V_h(T) = e^{\lambda T}$, by plugging in

$$2\frac{r}{2\rho w^2} \lambda^2 e^{\lambda T} + c\lambda e^{\lambda T} + g e^{\lambda T} = 0 \Rightarrow -\frac{r\lambda^2}{2\rho w^2} + c\lambda + g = 0$$

$$\lambda_\pm = \frac{\frac{2\rho w^2 c}{r} \pm \sqrt{\left(\frac{2\rho w^2 c}{r}\right)^2 + 4\frac{2\rho w^2 g}{r}}}{2}$$

$\lambda_\pm$ are real. $\lambda_+ > 0$ $\lambda_- < 0$

on $[-T,0]$ we use $e^{\lambda_-(T+T_0)}$ $e^{\lambda_+ T}$

on $[0,T_1]$ we use $e^{\lambda_- T}$ $e^{\lambda_+(T-T_1)}$

Now we consider the inhomogeneous terms

We solve $-\frac{r}{2\rho w^2} V''(T) + cV'(T) + gV(T) = RHS(T)$

We know $RHS(T)$ as $P + Qe^{\left(-\frac{T}{r}\right)}$

$V_p(T) = A + Be^{-T/r}$, we know $V(T) = A \Rightarrow A = \frac{P}{g}$

If $V(T) = Be^{-\frac{T}{r}}$ we have $LHS = Be^{-T/r}\left(g - \frac{c}{r} - \frac{r}{2\rho w^2 r^2}\right) \Rightarrow B = \frac{Q}{g - \frac{c}{r} - \frac{r}{2\rho w^2 r^2}}$

on $[-T_0, 0]$, $M = 0$

$RHS = -I_k N(T) + I_0 = I_0 - I_k N_1 e^{-T_0/r} e^{-T/r}$ by plugging in $N(T)$

$P_1 = I_0$ $Q_1 = -I_k N_1 e^{-T_0/r}$ we know $V_p^{(1)}(T) = \frac{I_0}{g} + \frac{-I_k N_1 e^{-T_0/r}}{g - \frac{c}{r} - \frac{r}{2\rho w^2 r^2}} e^{-T/r}$

on $(0, T_1)$ $M = 1$

$$\begin{aligned} RHS &= I_{Na}(1-N) - I_k N + I_0 \\ &= I_0 - I_k + (I_{Na} + I_k)(1-N_0) e^{-T/r} \end{aligned}$$

$P_2 - I_0 - I_k$ $Q_2 = (I_{Na} + I_k)(1-N_0)$

$$V_p^{(2)}(T) = \frac{I_0 - I_k}{g} + \frac{(I_{Na} + I_k)(1-N_0)}{g - \frac{c}{r} - \frac{r}{2\rho w^2 r^2}} e^{-T/r}$$

We know on $[-T_0,0]$ $V_1(T)=V_p^{(1)}(T)+a_1 e^{\lambda_-(T+T_0)}+b_1 e^{\lambda_+ T}$

on $(0,T_1)$ $V_2(T)=V_p^{(2)}(T)+a_2 e^{\lambda_- T}+b_2 e^{\lambda_+(T-T_1)}$

Similar as 2. we know $V(-T_0)=V(0)=V(T_1)=V^*$

on $[-T_0,0]$, $V_p^{(1)}(-T_0)+a_1+b_1 e^{\lambda_+(-T_0)}=V^* \Rightarrow a_1+b_1 e^{-\lambda_+ T_0}=V^*-V_p^{(1)}(-T_0)$

$$V_p^{(1)}(0)+a_1 e^{\lambda_- T_0}+b_1=V^* \Rightarrow a_1 e^{\lambda_- T_0}+b_1=V^*-V_p^{(1)}(0)$$

we obtained that $b_1=\dfrac{V^*-V_p^{(1)}(0)-(V^*-V_p^{(1)}(-T_0))e^{(\lambda_- T_0)}}{1-e^{(\lambda_- -\lambda_+)T_0}}$

$$a_1=V^*-V_p^{(1)}(-T_0)-b_1 e^{-\lambda_+ T_0}$$

on $(0,T_1)$ we have $V(0)=V(T_1)=V^*$

Similarly we have $\begin{cases} a_2+b_2 e^{-\lambda_+ T_1}=V^*-V_p^{(2)}(0)\\ a_2 e^{\lambda_- T_1}=V^*-V_p^{(2)}(T_1)\end{cases}$

we obtained $b_2=\dfrac{V^*-V_p^{(2)}(T_1-(V^*-V_p^{(2)}(0))e^{(\lambda_- T_1)}}{1-e^{(\lambda_- -\lambda_+)T_1}}$

$\cancel{V_1(T)=V_p^{(1)}(T)+a_1 e}$ $\lambda_\pm$ are the roots of characteristic Equation

$$-\frac{r}{2\rho w^2}\lambda^2+c\lambda+g=0$$

$$V_p^{(1)}(T)=\frac{I_0}{g}-\frac{I_k N_0}{\Delta}e^{-T/\tau},\quad V_p^{(2)}(T)=\frac{I_0-I_k}{g}+\frac{(I_{na}+I_k)(1-N_0)}{\Delta}e^{-T/\tau}$$

$$\Delta:=g-\frac{c}{\tau}-\frac{r}{2\rho w^2\tau^2}$$

Since the travelling-wave equation for V is second order, we must additionally impose derivative matching

$$V_1'(0)=V_2'(0^+)\qquad V_1'(-T_0)=V_2'(T_1)$$

These two determine (N_0,N_1) for each chosen (I_0,w) numerically, for fixed I_0 we solve the 2x2 nonlinear system in (N_0,N_1) and the compute the period $P=T_0+T_1$

3 Continue

Introduce the $\varepsilon = \frac{c/g}{\tau} = \frac{c}{g\tau}$ $w = \theta \frac{\sqrt{r/(\rho g)}}{(c/g)} = \theta \frac{\sqrt{r/(\rho g)}}{\varepsilon \tau}$

$$\lambda_{\pm} = O(\tfrac{1}{\varepsilon}) \quad \Delta = g(1+O(\varepsilon))$$

As $\varepsilon \to 0$, $b_1 \approx V^* - V_p^{(1)}(0)$, $a_1 \approx V^* - V_p^{(1)}(-T_0)$ $a_2 \approx V^* - V_p^{(2)}(0)$ $b_2 \approx V^* - V_p^{(2)}(T_1)$

The derivative matching then reduces to simple proportionality relations

$$a_2 = -\gamma b_1 \quad a_1 = -\gamma b_2$$

where $\gamma > 1$ $\theta = \frac{\sqrt{2\gamma}}{\gamma - 1}$

Let $P = T_0 + T_1$ $E = e^{P/\tau}$ and $A = I_0 - gV^*$. The $\varepsilon \to 0$ algebra yields the closed form relation $E = \frac{I_{na}\gamma + A(\gamma+1)}{I_{na} + A(\gamma+1)} \Rightarrow \gamma = \frac{EI_{na} + EA - A}{I_{na} + A - EA}$.

Therefore, given the period P, we compute $\gamma(P)$, then $\theta(P) = \frac{\sqrt{2\gamma}}{\gamma - 1}$.

We finally solved the wave speed $w(P) = \theta(P) = \frac{\sqrt{r/(\rho g)}}{\varepsilon \tau}$